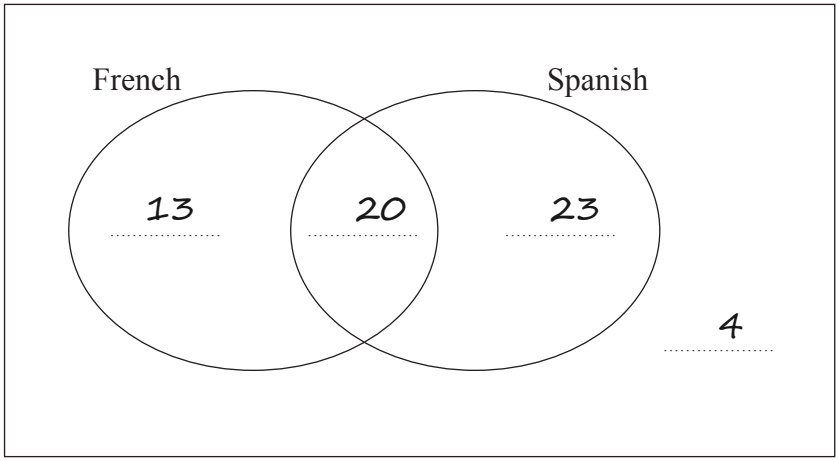
23 There are 60 students at a college.

20 students study both French and Spanish.
13 students study French but not Spanish.
A total of 43 students study Spanish.

(a) Complete the Venn diagram for this information.



1 mark
13 and 20 placed correctly

1 mark
Subtract known counts

1 mark
Complete diagram

(3)

Spanish only: 43 – 20 = 23; neither: 60 – 13 – 20 – 23 = 4.

One of the students at the college is to be selected at random.

(b) Write down the probability that this student studies neither French nor Spanish.

$\frac{4}{60} = \frac{1}{15}$

1 mark
4/60 = 1/15

(1)

(Total for Question 23 is 4 marks)